

Week 18: SELinux & Security Modules

Date: July 6, 2026

Study Time: 5-8 hours total

GOAL

Understand SELinux basics and troubleshoot common issues

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" - Chapter 17 (SELinux section)

Video Learning

- ☐ LearnLinuxTV - "SELinux Tutorial"
- ☐ Professor Messer - "SELinux"

Concepts

- ☐ What is SELinux?
 - ☐ Mandatory Access Control (MAC) vs Discretionary Access Control (DAC)
 - ☐ SELinux modes: Enforcing, Permissive, Disabled
 - ☐ Security contexts and labels
 - ☐ SELinux booleans
 - ☐ SELinux policies (targeted, minimum, mls)
 - ☐ File contexts
 - ☐ Process contexts
-

LAB WORK (3-4 hours)

Understanding SELinux Status

Note: This week is primarily for your **Fedora server**, as Fedora uses SELinux by default. Ubuntu uses AppArmor instead.

- ☐ Check if SELinux is installed

```
bash
```

```
ssh fedora
```

```
# Check SELinux status
```

```
sestatus
```

```
# Should show:
```

```
# SELinux status: enabled
```

```
# Current mode: enforcing
```

```
# Policy: targeted
```

☐ Check current mode

```
bash
```

```
getenforce
```

```
# Returns: Enforcing, Permissive, or Disabled
```

☐ Understand the modes

```
Enforcing - SELinux is active and blocking
```

```
Permissive - SELinux logs violations but doesn't block
```

```
Disabled - SELinux is completely off
```

SELinux Contexts

Everything in SELinux has a context (label).

☐ View file context

```
bash
```

```
ls -Z /var/www/html/
```

```
# Shows contexts like:
```

```
# -rw-r--r--. root root unconfined_u:object_r:httpd_sys_content_t:s0 index.html
```

```
# Context format: user:role:type:level
```

☐ View your own process context

bash

```
id -Z  
# Shows your SELinux context
```

☐ View running processes contexts

bash

```
ps auxZ  
# Shows SELinux context for each process  
  
# Or specific process:  
ps -eZ | grep sshd
```

☐ View directory context

bash

```
ls -ldZ /var/www/  
ls -ldZ /home/  
ls -ldZ /var/lib/gitea/
```

Temporarily Change SELinux Mode

☐ Switch to Permissive mode (for testing)

bash

```
# Temporarily (until reboot):  
sudo setenforce 0  
  
# Verify  
getenforce # Should show: Permissive
```

☐ Switch back to Enforcing

bash

```
sudo setenforce 1  
  
# Verify  
getenforce # Should show: Enforcing
```

☐ Permanent mode change (requires reboot)

```
bash
```

```
sudo nano /etc/selinux/config
```

```
# Change this line:
```

```
SELINUX=enforcing
```

```
# To:
```

```
SELINUX=permissive
```

```
# Or:
```

```
SELINUX=disabled
```

```
# Save and reboot
```

```
sudo reboot
```

Common SELinux Scenario: Gitea

Let's troubleshoot a typical SELinux issue with your Gitea installation.

☐ Check Gitea file contexts

```
bash
```

```
ls -lZ /var/lib/gitea/
```

```
ls -lZ /etc/gitea/
```

☐ If Gitea isn't working, check for denials

```
bash
```

```
# View recent SELinux denials
```

```
sudo ausearch -m avc -ts recent
```

```
# Or check audit log
```

```
sudo tail -100 /var/log/audit/audit.log | grep denied
```

☐ Common Gitea context fix

```
bash
```

```
# Set correct context for Gitea data
```

```
sudo semanage fcontext -a -t httpd_sys_rw_content_t "/var/lib/gitea(/.*)?"
```

```
# Apply the context
```

```
sudo restorecon -Rv /var/lib/gitea/
```

Fixing File Contexts

☐ Temporarily change context (chcon - doesn't survive relabel)

bash

```
# Example: Allow Apache to read a file
sudo chcon -t httpd_sys_content_t /path/to/file

# Verify
ls -Z /path/to/file
```

☐ Permanently change context (semanage + restorecon)

bash

```
# Add rule
sudo semanage fcontext -a -t httpd_sys_content_t "/custom/web/path(/.*)"

# Apply rule
sudo restorecon -Rv /custom/web/path/
```

☐ Restore default contexts

bash

```
# Restore one file
sudo restorecon -v /path/to/file

# Restore directory recursively
sudo restorecon -Rv /var/www/html/
```

SELinux Booleans

Booleans are on/off switches for specific permissions.

☐ List all booleans

bash

```
getsebool -a
# Shows hundreds of booleans

# Search for specific ones
getsebool -a | grep httpd
getsebool -a | grep ssh
```

☐ Check specific boolean

bash

```
getsebool httpd_can_network_connect
# Shows: on or off
```

☐ Set boolean temporarily

bash

```
sudo setsebool httpd_can_network_connect on
# Changes until reboot
```

☐ Set boolean permanently

bash

```
sudo setsebool -P httpd_can_network_connect on
# -P makes it persistent across reboots
```

☐ Common booleans you might need

bash

```
# Allow httpd to connect to network
sudo setsebool -P httpd_can_network_connect on

# Allow httpd to send email
sudo setsebool -P httpd_can_sendmail on

# Allow SSH to use TCP forwarding
sudo setsebool -P ssh_sysadm_login on
```

Troubleshooting SELinux Issues

☐ Test if SELinux is causing the problem

bash

```
# Temporarily disable SELinux
sudo setenforce 0

# Try your application
# Does it work now?

# If YES: SELinux is the problem
# If NO: Problem is elsewhere

# Re-enable SELinux
sudo setenforce 1
```

☐ View recent denials

bash

```
sudo ausearch -m avc -ts recent

# More readable format
sudo ausearch -m avc -ts recent | audit2why
```

☐ Get suggested fixes

bash

```
# Install troubleshooting tools
sudo dnf install setroubleshoot-server

# View suggestions
sudo sealert -a /var/log/audit/audit.log

# This shows:
# - What was denied
# - Why it was denied
# - Suggested fixes
```

Practical Exercise: Web Server

☐ Install Apache (for testing)

bash

```
sudo dnf install httpd
sudo systemctl start httpd
sudo systemctl enable httpd

# Open firewall
sudo firewall-cmd --permanent --add-service=http
sudo firewall-cmd --reload
```

☐ Create test file in wrong location

bash

```
# Create file in /root (wrong!)
sudo touch /root/test.html
echo "<h1>Test Page</h1>" | sudo tee /root/test.html

# Try to serve it (this will fail!)
sudo ln -s /root/test.html /var/www/html/test.html

# Try to access: http://your-fedora-ip/test.html
# Should get 403 Forbidden
```

☐ Check SELinux denial

bash

```
sudo ausearch -m avc -ts recent
# Should show httpd was denied access
```

☐ Fix with correct context


```
bash
```

```
# Move file to correct location
sudo mv /root/test.html /var/www/html/

# Set correct context
sudo restorecon -v /var/www/html/test.html

# Verify context
ls -Z /var/www/html/test.html
# Should show: httpd_sys_content_t

# Now it should work!
```

SELinux Policy Queries

☐ List all types

```
bash
```

```
seinfo -t
# Shows all SELinux types
```

☐ Find what a process can do

```
bash
```

```
sesearch -A -s httpd_t
# Shows all allowed actions for httpd
```

☐ Find what can access a type

```
bash
```

```
sesearch -A -t httpd_sys_content_t
# Shows what can access httpd content
```

Create SELinux Troubleshooting Script

☐ SELinux status script

bash

```
nano ~/selinux-status.sh

#!/bin/bash

echo "=== SELinux Status Report ==="
echo "Generated: $(date)"
echo ""

echo "--- Current Status ---"
sestatus
echo ""

echo "--- Current Mode ---"
getenforce
echo ""

echo "--- Recent Denials (last 10) ---"
sudo ausearch -m avc -ts recent | tail -20
echo ""

echo "--- Common Boolean Settings ---"
getsebool httpd_can_network_connect
getsebool httpd_can_sendmail
getsebool ssh_sysadm_login
echo ""

echo "--- Custom File Contexts ---"
sudo semanage fcontext -l | grep -v "system_u:object_r:default_t"

chmod +x ~/selinux-status.sh
```

Document Your SELinux Configuration

☐ Create SELinux documentation

bash

```
nano ~/selinux-config.md
```

```
# SELinux Configuration Documentation
```

```
## Current Mode
```

- Mode: Enforcing/Permissive/Disabled
- Policy: targeted

```
## Custom File Contexts
```

- /var/lib/gitea: httpd_sys_rw_content_t
- [Add others]

```
## Booleans Modified
```

- httpd_can_network_connect: on (Reason: Gitea needs network)
- [Add others]

```
## Known Issues & Solutions
```

1. Gitea can't write to data directory
 - Solution: semanage fcontext + restorecon

```
## Troubleshooting Commands
```

- Check denials: `sudo ausearch -m avc -ts recent`
- Get suggestions: `sudo sealert -a /var/log/audit/audit.log`
- Temp disable: `sudo setenforce 0`

FLASHCARD COMMANDS

1. `getenforce`
2. `setenforce`
3. `sestatus`
4. `ls -Z`
5. `ps -Z`
6. `chcon`
7. `restorecon`
8. `semanage fcontext`
9. `getsebool`
10. `setsebool`
11. `ausearch`
12. `sealert`
13. `audit2why`

Concepts:

- MAC vs DAC?
- Three SELinux modes?
- What is a context?
- What is a boolean?
- chcon vs semanage?

EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

QUARTERLY CHECK-IN #3

Message me: "Quiz me on Weeks 13-18"

Topics covered:

- Bash scripting (functions, arrays, cron)
- Package management (apt, dnf, rpm)
- Security scanning (nmap, lynis)
- SELinux basics

Self-assessment:

- Confidence level (1-10): _____
 - Topics needing review: _____
-



NOTES

SELinux mode on Fedora:

- Current: _____
- Changed to: _____

Issues encountered:

Contexts modified:

Booleans changed:

Completed: ☐



QUICK REFERENCE

SELinux Modes:

```
bash

getenforce          # Check current mode
sudo setenforce 0   # Permissive (temporary)
sudo setenforce 1   # Enforcing (temporary)
```

File Contexts:

```
bash
```

```
ls -Z file # View context
sudo chcon -t TYPE file # Temp change
sudo restorecon -v file # Restore default
sudo semanage fcontext -a -t TYPE "path(/.*)?" # Add rule
sudo restorecon -Rv path # Apply rule
```

Booleans:

```
bash
```

```
getsebool -a # List all
getsebool BOOL # Check one
sudo setsebool BOOL on # Temp change
sudo setsebool -P BOOL on # Permanent
```

Troubleshooting:

```
bash
```

```
sudo ausearch -m avc -ts recent # Recent denials
sudo sealert -a /var/log/audit/audit.log # Suggestions
sudo setenforce 0 # Temp disable to test
```

Common Contexts:

- httpd_sys_content_t - Web content (read-only)
- httpd_sys_rw_content_t - Web content (read-write)
- user_home_t - User home directories
- tmp_t - Temporary files
- etc_t - Configuration files

Previous Week: Security Scanning

Next Week: Storage & Filesystems (Part 1)